CBC SppGrasp: evidence for an adaptive default

564 validated solves on 47 SPP-dominant models

28 October 2026

Executive result

The tested 500M-work candidate lowers whole-collection mean PDI by **29.85% versus OFF**, preserves **every one of OFF's 104 optimal solves**, and reduces runs with no incumbent from **16 to 4**. It recovers both proofs lost by unchanged ON on `eilC76-2` (seeds 6401 and 6402). There are no proof gains or losses versus OFF.

This is not a regression-free or universal-dominance claim. Mean runtime is 839.85 s versus 839.42 s for OFF (+0.43 s); individual PDI and final-gap regressions remain. The candidate matches, rather than exceeds, OFF's optimal count. No AUTO mode or default change has been implemented by this experiment.

Whole-collection results

Metric	OFF	Unchanged ON	500M + FPump retained
Validated optimal solves / 188	104	102	104
Mean PDI (lower is better)	0.285123	0.201818	0.200023
Mean elapsed seconds	839.42	849.18	839.85
No incumbent at 1,800 s	16	4	4
Mean final gap %, same 172 pairs	8.832	8.522	8.275

Paired mean PDI difference, candidate minus OFF: **-0.085100**; descriptive 95% instance-bootstrap interval **[-0.166489, -0.022779]**. All seeds of each instance are resampled together (10,000 samples). This interval is conditional on the selected collection, not an estimate for unseen MIPs.

What was tested

The collection contains **47 SPP-dominant instances**: 29 pure binary set packing/partitioning models and 18 separately labeled supplemental RMP application models. Seventeen are from MIPLIB 2017; 30 are collected application instances. At least 90% of original rows must be packing or partitioning (actual minimum 92.044%). The feature table and inputs were frozen before the original performance study.

All 47 models were run in all three arms with four matched seeds: **6401, 6402, 6601, 6602**. Each solve received **1,800 elapsed seconds**; 16 independent serial CBC processes (`-threads 0`) ran concurrently, with BLAS/OpenMP restricted to one thread. Task order was shuffled with scheduling seed 6500. All **564 planned runs validated**.

Arm	Explicit settings
OFF	`-sppGrasp off`
Unchanged ON	`-sppGrasp on`
Candidate	`-sppGrasp on -sppGraspWork 500 -sppGraspFpump on`

`sppGraspWork 500` is 500 million counted work units per call, not 500 seconds or a hard whole-solve work budget. `sppGraspFpump on` prevents automatic disabling; it does not force FPump to perform LP passes when an incumbent already exists. Existing applicability, native completion and all other search defaults remain in effect.

The arms use the **same frozen optimized, Nauty-free executable** and dependencies. No model-name gate, input-model modification, forced applicability or per-instance parameter tuning was applied to the candidate. Existing automatic artificial-slack behavior is retained.

PDI and solution validation

PDI is the average normalized primal-dual gap over the common 1,800-second horizon. Gap is 2 without an incumbent, 1 with an incumbent but no usable dual bound (or opposite signs), otherwise `abs(primal-dual)/max(abs(primal),abs(dual))`, with the analyzer's zero-gap convention for equal zero bounds. Lower is better. An optimality/infeasibility proof closes the gap at its actual timestamp; a stopped run holds its last gap to the horizon.

Every history uses the complete progress CSV, including original-model incumbents and a positive global final timestamp. No missing CSV or fractional LP relaxation is converted into an integer solution. An independent validator checks genuine incumbents against bounds, integrality, every row and objective; reference objectives cross-check proof claims. Failed or missing runs block complete reporting.

Conventional final relative gap uses `abs(p-d)/max(abs(p),1e-10)` and is averaged only over the same 172 finite-bound/incumbent pairs in all arms. It differs from the normalized PDI gap.

Finding solutions and proving optimality over time

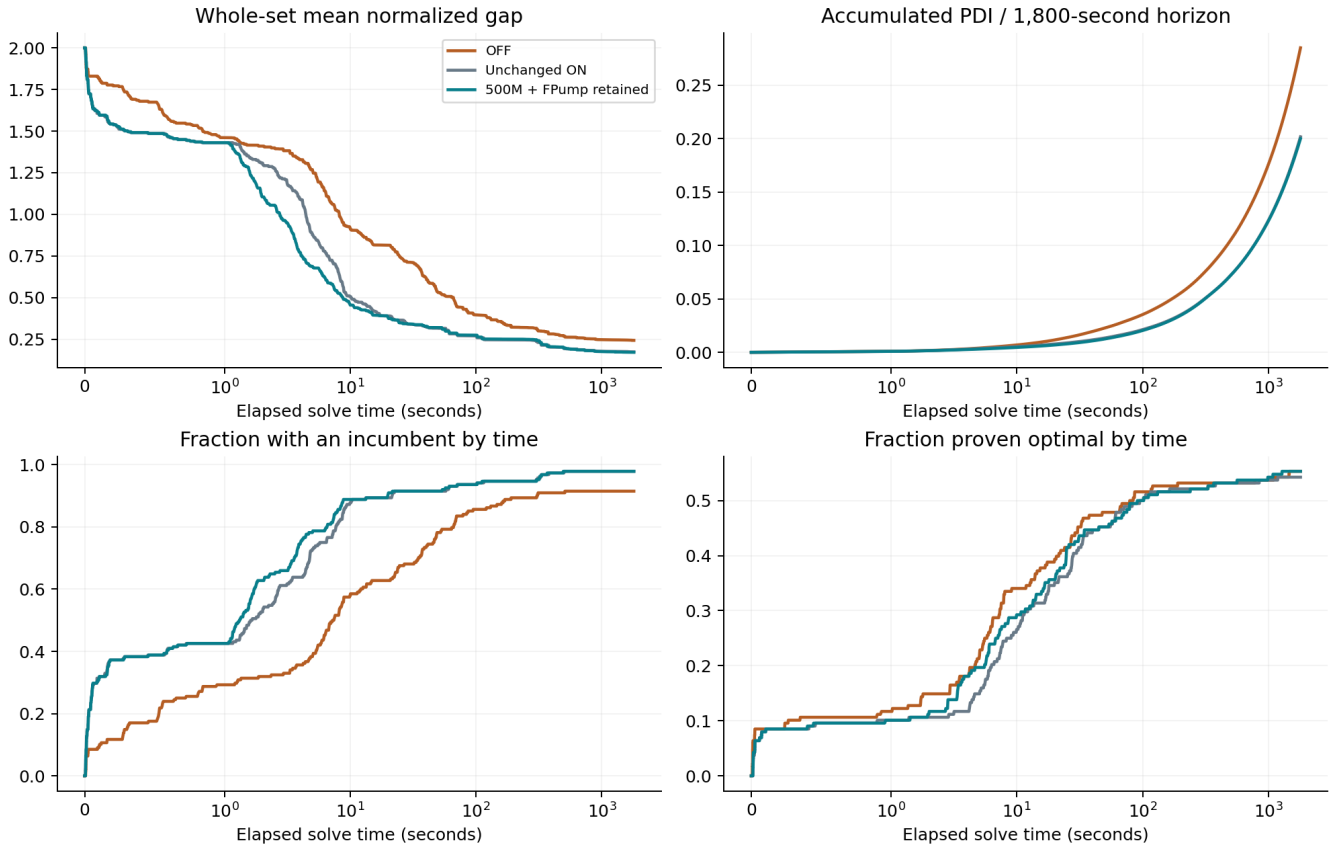


Figure 1: All 188 runs per arm; the final cumulative ordinate equals the reported mean PDI. Time uses a symmetric log scale near zero.

Original and fresh seeds

Original seeds deliberately revisit the earlier proof losses. Fresh seeds provide seed-based replication on the same models, not an independent-instance holdout.

Original seeds

Arm	Optimal / 94	Mean PDI	Mean seconds	No incumbent
OFF	52	0.283897	832.57	8
Unchanged ON	50	0.203019	857.56	2
500M + FPump retained	52	0.200525	836.52	2

Fresh seeds

Arm	Optimal / 94	Mean PDI	Mean seconds	No incumbent
OFF	52	0.286349	846.26	8
Unchanged ON	52	0.200617	840.80	2
500M + FPump retained	52	0.199521	843.18	2

Individual-model effects

Each bar averages four matched seeds; negative favors the candidate. Regressions are retained, not omitted.

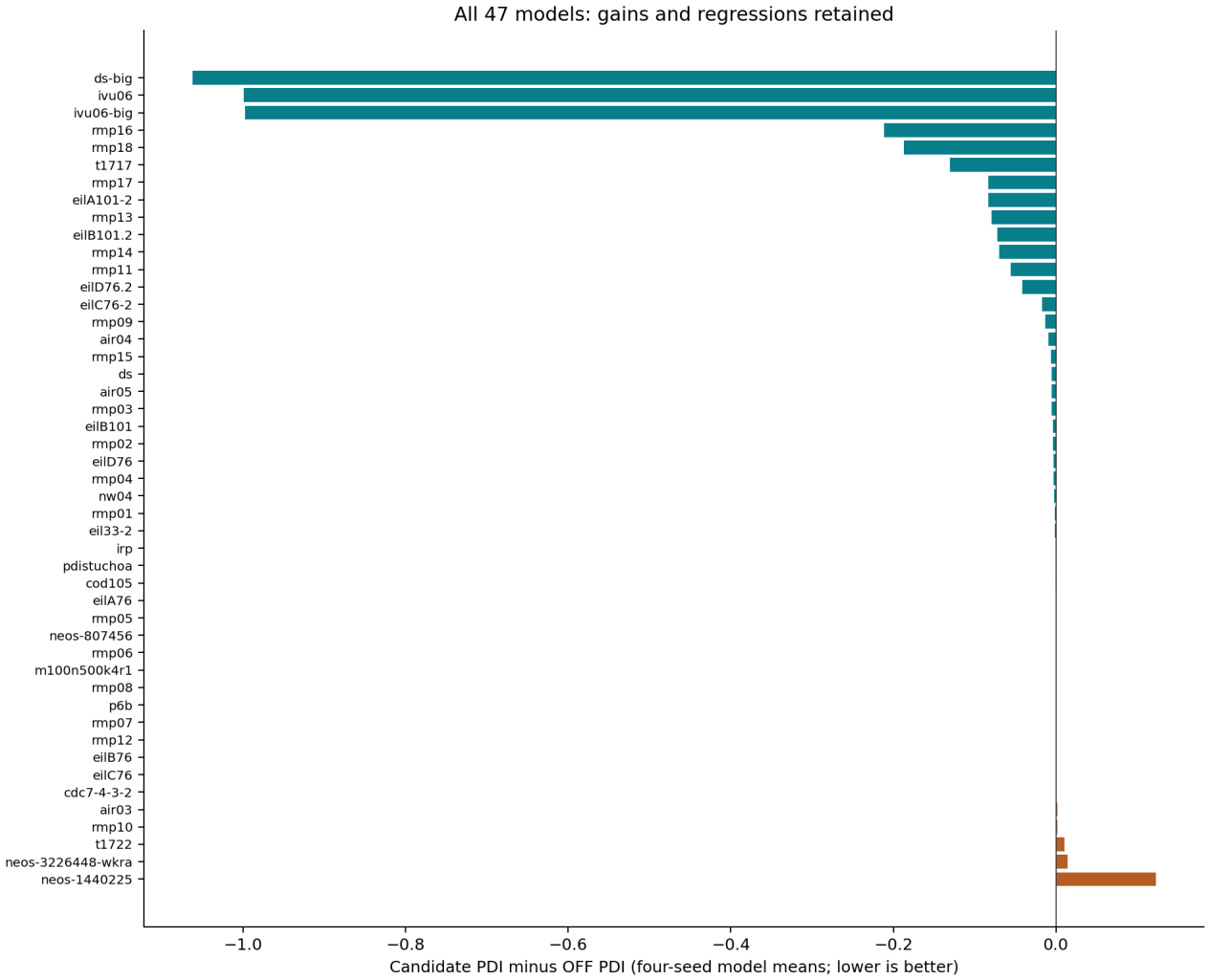


Figure 2: Candidate versus OFF on every model.

Interpretation for AUTO

Forrest's concern is valid: AIR03 profiling exposed expensive local search, and his 44-problem unit-test timing increased from about 85.4 s to 95.3 s when SppGrasp was enabled. This study does not refute that measurement or establish that a broad default should always execute full SppGrasp.

The evidence supports an **AUTO policy proposal**, not simply renaming unchanged ON. The tested starting point is existing structural applicability plus a 500M per-call work limit and preservation of FPump availability, while retaining safe pre-LP native completion. A general AUTO implementation still needs explicit semantics, user-override preservation, and broader non-SPP/quick-suite validation. A cheap structural skip and effort control are hypotheses for that implementation, not separately proven adaptive decisions here.

The 500M candidate was selected from a seven-policy screen on 11 models. The confirmation uses all 47 models and fresh seeds but the collection has tuning/development exposure. Do not describe it as unseen-instance validation, or assert a guarantee of no regressions. Timing under concurrent load is noisy; the 0.43 s aggregate runtime difference is reported, not claimed statistically equivalent.

Reporting fixes and reproducibility

A previous confirmation finished with only 562/564 valid runs: an OFF `ivu06/6602` preprocessing timeout omitted its CSV, while an ON timeout reported a saved incumbent but exported the fractional LP vector. Both original failures remain preserved. A reporting-only fix restores the saved integer vector and finalizes progress on that early return. The full 564-run round was restarted with the same repaired binary in all arms; no old/new results were mixed.

Baseline: `coin-or/Cbc@97ca70c7` (master, OFF default). Reporting-only source: `f7ef1ae9` (full revision in the manifest), including initial-LP and preprocessing-timeout repairs. It does not import unrelated next heuristic changes. Focused timeout probes fail on the old build and pass after the repair; native/progress/objective-offset, solver quick, callback and bound tests passed. Nauty was excluded from every arm.

The result directory includes the exact manifest, all source revisions, commands, input/binary/runtime-library hashes, combined source patch, per-run metrics, proof transitions and gap regressions. Raw evidence and frozen executables accompany the experiment release.

References

- Dataset, model characteristics and attribution
- Forrest's OFF-default commit and discussion
- Gleixner et al. (2021), *MIPLIB 2017: Data-Driven Compilation of the 6th Mixed-Integer Programming Library*, Mathematical Programming Computation. DOI: 10.1007/s12532-020-00194-3.

Appendix: all model means and optimal counts

Optimal counts are out of four seeds per model and arm. No proof gain or loss occurs between the candidate and OFF.

Model	OFF PDI	ON PDI	Candidate PDI	OFF/ON/Candidate optima
air03	0.001097	0.007485	0.002695	4 / 4 / 4
air04	0.015436	0.007268	0.006034	4 / 4 / 4
air05	0.008120	0.005210	0.002811	4 / 4 / 4
cdc7-4-3-2	0.815156	0.814714	0.815628	0 / 0 / 0
cod105	0.319143	0.318328	0.318835	0 / 0 / 0
ds	0.838676	0.835127	0.833208	0 / 0 / 0
ds-big	2.000000	0.948118	0.937527	0 / 0 / 0
eil33-2	0.004001	0.004967	0.003043	4 / 4 / 4
eilA101-2	0.279364	0.251301	0.196173	0 / 0 / 0
eilA76	0.001336	0.002345	0.001306	4 / 4 / 4
eilB101	0.008565	0.006972	0.004899	4 / 4 / 4
eilB101.2	0.180495	0.109614	0.108445	0 / 0 / 0
eilB76	0.000660	0.000815	0.000813	4 / 4 / 4
eilC76	0.001528	0.002396	0.001733	4 / 4 / 4
eilC76-2	0.043454	0.042206	0.026591	4 / 2 / 4
eilD76	0.006855	0.005262	0.003908	4 / 4 / 4
eilD76.2	0.126526	0.087587	0.085144	0 / 0 / 0
irp	0.001675	0.002514	0.001041	4 / 4 / 4
ivu06	2.000000	1.000758	1.000771	0 / 0 / 0
ivu06-big	2.000000	1.002550	1.002512	0 / 0 / 0
m100n500k4r1	0.040350	0.040349	0.040353	0 / 0 / 0
neos-1440225	0.116972	0.245932	0.240157	4 / 4 / 4
neos-3226448-wkra	0.066103	0.081044	0.080806	4 / 4 / 4
neos-807456	2.000000	2.000000	2.000000	0 / 0 / 0
nw04	0.003756	0.003863	0.001819	4 / 4 / 4
p6b	0.076293	0.076282	0.076310	0 / 0 / 0
pdistuchoa	0.230456	0.230210	0.230041	0 / 0 / 0
rmp01	0.003872	0.003262	0.002645	4 / 4 / 4
rmp02	0.006235	0.002993	0.002718	4 / 4 / 4
rmp03	0.007347	0.002759	0.002218	4 / 4 / 4
rmp04	0.005832	0.003557	0.002962	4 / 4 / 4
rmp05	0.000012	0.000014	0.000012	4 / 4 / 4
rmp06	0.000008	0.000011	0.000011	4 / 4 / 4
rmp07	0.000027	0.000049	0.000051	4 / 4 / 4
rmp08	0.000011	0.000020	0.000020	4 / 4 / 4
rmp09	0.015976	0.004265	0.003224	4 / 4 / 4
rmp10	0.005764	0.009102	0.007828	4 / 4 / 4
rmp11	0.155552	0.093342	0.099812	0 / 0 / 0
rmp12	0.000120	0.000157	0.000164	4 / 4 / 4
rmp13	0.086616	0.009309	0.007300	0 / 0 / 0
rmp14	0.083730	0.016771	0.014318	0 / 0 / 0
rmp15	0.009691	0.003481	0.003521	4 / 4 / 4
rmp16	0.581765	0.372977	0.370370	0 / 0 / 0
rmp17	0.286253	0.203952	0.202846	0 / 0 / 0
rmp18	0.279623	0.060156	0.092273	0 / 0 / 0
t1717	0.454049	0.328746	0.323399	0 / 0 / 0
t1722	0.232279	0.237299	0.242791	0 / 0 / 0

Appendix: terminal-gap regressions

Retained for transparency, even though final gap is secondary to PDI, optimal solves and solution availability in this study.

Candidate versus OFF

- cod105/6401: 33.61257% to 33.62634%.
- cod105/6402: 33.50752% to 33.53479%.
- ds/6401: 70.10091% to 81.32828%.
- ds/6402: 73.82693% to 82.44223%.
- ds/6602: 75.34253% to 91.98872%.
- eilA101-2/6601: 12.88198% to 13.39127%.
- eilA101-2/6602: 13.36729% to 14.19117%.
- eilB101.2/6401: 7.01992% to 13.56656%.
- eilB101.2/6402: 7.93236% to 11.30707%.
- rmp11/6401: 6.11115% to 10.47024%.
- rmp11/6402: 5.27074% to 9.71279%.
- rmp11/6601: 7.75972% to 8.54651%.
- rmp13/6401: 0.02061% to 0.03159%.
- rmp13/6402: 0.04688% to 0.06537%.
- rmp13/6602: 0.03003% to 0.04462%.
- rmp14/6601: 0.29069% to 0.36632%.
- rmp14/6602: 0.27479% to 0.46336%.
- rmp18/6402: 0.50635% to 0.75110%.
- rmp18/6601: 0.59792% to 0.69245%.
- t1717/6401: 29.49143% to 31.08301%.
- t1717/6402: 30.12918% to 32.07847%.
- t1722/6401: 21.67540% to 23.30679%.
- t1722/6601: 23.10517% to 23.67526%.
- t1722/6602: 14.51908% to 26.98501%.

Candidate versus ON

- cod105/6401: 33.57171% to 33.62634%.
- cod105/6601: 33.45654% to 33.48377%.
- ds/6602: 73.41767% to 91.98872%.
- ds-big/6402: 87.75136% to 94.39318%.
- eilB101.2/6401: 5.97076% to 13.56656%.
- eilB101.2/6601: 3.04014% to 5.53413%.
- eilD76.2/6402: 5.25366% to 5.29881%.
- pdistuchoa/6402: 23.92484% to 23.95382%.
- rmp11/6402: 8.92147% to 9.71279%.
- rmp11/6601: 7.73265% to 8.54651%.
- rmp11/6602: 6.85100% to 8.98772%.
- rmp13/6602: 0.03125% to 0.04462%.
- rmp14/6601: 0.33342% to 0.36632%.
- rmp14/6602: 0.25866% to 0.46336%.
- rmp16/6601: 33.72837% to 35.02142%.
- rmp16/6602: 36.97225% to 37.57182%.
- rmp18/6402: 0.52400% to 0.75110%.
- rmp18/6601: 0.40676% to 0.69245%.
- rmp18/6602: 0.61212% to 0.65704%.
- t1717/6402: 31.12962% to 32.07847%.
- t1717/6601: 30.80197% to 30.94173%.
- t1717/6602: 31.92736% to 32.32027%.

- t1722/6401: 21.08908% to 23.30679%.
- t1722/6601: 20.64669% to 23.67526%.
- t1722/6602: 21.60980% to 26.98501%.